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Speech recognition with principal component analysis

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Abstract

Various researches on speech recognition are carried out nowadays; moreover, hardware and software are developed on this subject. Operations can easily and rapidly be carried out by establishing interaction between computer and human via verbal communication. Therefore, studies related to speech recognition remain important today. Various algorithms have been used in speech recognition processes. Principal Component Analysis was utilised in the recognition of verbal statements within the scope of this study. Initially, eigenvalues and eigenvectors of 100 different words recorded in computer settings were obtained; afterwards, recognition of verbal statement stored in the system is performed by comparing the eigenvectors of verbal statements. The performance and usability of Principal Component Analysis in the recognition of verbal statements and voice commands is examined within the settings of this study. At the experimental practices, it is seen that significant results were obtained in speech recognition by using Principal Component Analysis method.

Keywords: Verbal Statement Recognition, Speech Recognition, Principal Component Analysis

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1. Introduction

Various methods have been being utilized within the communication between humans and machines besides the developments seen in technology. Speech recognition is one of the methods developed in order to establish communication between humans and machines. Humans begin to communicate through speaking before reading and writing. Therefore, speaking is one of the frequent communication method utilized in daily life.

The topic of performing various tasks utilizing voice commands in personal computer has been one of the subjects being studied nowadays. Computers that can display user's speech in a reliable way or perform the

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tasks according to the user's voice commands make various operations carried out in many areas easier. Dictation packages, voice browsing, automatic systems based on speech recognition via telephones, systems providing various alternatives to visually, physically, etc. handicapped individuals in order to communicate with machines can be given as examples. Voice web browsing systems are another applications used nowadays. Web browsing is realized over telephone directly with these types of systems that are not possible at the past. Operation such as banking reservations via telephone is one of the common applications within the field of speech recognition. Operations carried out by using telephone buttons which take much time previously become quicker, cheaper and easier with the aid of speech recognition based systems. Mobile phones that have voice dial features and household goods that are reactive to voices are the known devices mentioned above that make our lives easier. When the studies related to speech recognition is taken into consideration, it is seen that increasing accuracy or validity of the systems and removing the dependencies (speaker, language, environment where speaking takes place) are the most important topics. It is apparent that more practical and important systems will occur in the future with the support of increasing accuracy or validity of the systems and removing the dependencies. [1,2,3]

When the studies related to speech recognition is chronologically taken into consideration, it is seen that the first electronic speech synthesizer titled as Voder and produced by AT&T Bell Labs is the earliest study (Dudley, Riezs and Watkins). Many academic studies were carried out on voice and speech recognition. SUR (Speech Understanding Research) was established by DARPA (Defense Advanced Research Projects Agency) in 1971 in order to develop a computer system which can recognize speeches. Popular toy "Speak and Spell" was developed by Texas Instruments in 1978. A speech chip was designed for "Speak and Spell" and this chip provides opportunities to take giant steps in the field of speech synthesis (more natural and close to human speech). A company titled Speech Works was established in 1984 and produced automatic speech recognition (ASR) systems over telephones. Word based dictation software developed firstly by Dragon Systems was exposed for sale in a market in 1995. Afterwards similar software was begun to be produced also by IBM and Kurzweil. [1,2]

In the last 15-20 years, various verbal statement recognition strategies were tried to be realized in different languages and forms peculiar to these languages. Speech recognition is related to various scientific areas such as signal processing, pattern recognition, artificial intelligence, information and probability theories, statistics, computer sciences, psychology, linguistics, biology, phonology [1,3,4].

Benzeghiba et al. [5] were found out that accent, gender, age and effects of diction vary according to climate and regions in speech recognition. Valente [6] was found out that hierarchical speech recognition was more effective compared to continuous speech recognition within the speech recognition study carried out in 2009. Kim and Stern [7] studied Multi-band method in order to minimise noises on speech recognition in their study carried out in 2010. Cooke et al. [8] test algorithms developed considering differentiating and removing mono sounds at the background and evaluate their results in their study they carried out on speech recognition in 2010. Kingsbury et al. [9] made a study on an effective speech recognition system whose noises removed via utilizing Modulation spectrogram in 1998. Fu and Juang [10] made study on automatic speech recognition system based on Bayes Classification and Weighted Minimum Classification error method. Wan and Lee [11] developed speech recognition system by utilizing Histogram Based Quantization method aiming to minimise environmental noises and filter peripheral noises on speech recognition in 2008. Kinjo and Funaki [12] made voice analysis by utilizing Hidden Markov Model and Mel-Frequency Filtering within complex sound setting in 2006.

There is no need of extra hardware for the processes of speech recognition or in other terms verbal statements (words) recognition. A microphone of high quality and necessary loaded software are adequate for this process. The subsequent step is to analyse voice data recorded through microphone with appropriate methods.

General purpose speech recognition software that can be used for security issues was developed by utilizing Principal Component Analysis (PCA) that is generally known and time performance analyses of PCA algorithm on speech recognition were done within the scope of this study.

2. PCA Algorithm

Principal Component Analysis (PCA) is used in the reduction of high dimensional data into smaller dimensions by taking distinctive characteristics into consideration [13, 14].

This procedure consists of two stages: 1) training stage and 2) recognition stage. In training stage, eigenmatrix which is required for recognition is formed from sample data for recognition.

One dimensional column matrix is formed from gray coloured 2 dimensional images. If we denote the images included in training set as $\Gamma_1, \Gamma_2, \ldots, \Gamma_M$ the mean related to these images is computed as follows:

$$\Psi = \frac{1}{M}\sum_{n=1}^{M}\Gamma_n \qquad (1)$$

The difference of each image from mean is obtained as below:

$$\Phi_i = \Gamma_i - \Psi \qquad (2)$$

The dimension of this high dimensional matrix requires to be reduced by finding orthonormal u_k vector in M number which denotes this high dimensional matrix optimally. Vectors are selected to maximise λ_k.

$$\lambda_k = \frac{1}{M}\sum_{n=1}^{M}(u_k^T \Phi_n) \qquad (3)$$

These λ_k values and u_k vectors are the eigenvalues and eigenvectors of covariance matrix.

$$C = \frac{1}{M}\sum_{n=1}^{M}(\Phi_n \Phi_n^T) \qquad (4)$$

$$C = AA^T$$

C matrix is in N^2 x N^2 dimension for the samples which are in N x N dimensions. Since it will be difficult to calculate the eigenvalues and eigenvectors of this matrix which is in these dimensions, the eigenvalues and eigenvectors of the matrix given below are calculated.

$$L = A^T A \qquad (5)$$

It is assumed that the number of the pictures will be smaller than the image dimension. Besides, eigenvalues of matrix which is in M x M dimension are adequate for the problem of face recognition. A new matrix consists of eigenvectors of $V = [v_1, v_2, v_3, \ldots., v_n]$ C covariance matrix and $A = [\Phi_1, \Phi_2, \Phi_3, \ldots., \Phi_n]$ which symbolises matrix consists of vectors displaying the differences of each face images from mean related to face and also which is denoted with $U = [u_1, u_2, u_3, \ldots., u_N]^T$ and eigenface matrix including N number eigenfaces is formed.

$$U = V.A^T \qquad (6)$$

Feature vector related to any face using eigenfaces is represented as follows:

$$w_k = u_k(\Gamma - \Psi) \qquad (7)$$

The image which will be used in the recognition stage should be in grayscale. Initially, the difference of Γ_T vector from mean related to face data is calculated as in Equation 8.

$$\Phi_T = \Gamma_T - \Psi \qquad (8)$$

In this stage, $w_T$feature vector of the face which will be recognised is calculated by multiplying Φ_T difference vector with $U = [u_1, u_2, u_3, ..., u_N]^T$ eigenface vector.

$$w_t = U.\Phi_T \tag{9}$$

Distance of calculated w_T feature vector to w_i feature vectors forming $W = [w_1, w_2, w_3, ..., w_N]$ matrix display the similarity of the image which will be processed to the face image recorded and which was being compared to feature vector. The response which will be generated by the system as recognised face is the image$_i$ which generates the lowest value out of d_i values calculated in Equation 10.

$$d_i = \sum_{j=1}^{M} |w_T(j) - w_i(j)|, i = 1,2,...N \tag{10}$$

3. Software

Various pre-processes were performed on voice data utilized in the development of speech recognition system. The first one among of all is sampling process. Sampling process is the digitisation of voice signal recorded as analogue by sampling it in specific time intervals. The size of amplitude of voice signal and the removal of noises from voice increase the performance of speech recognition process. Voice signals were sampled as 16bit and normalised at [0 1] interval utilising Equation 11 in this study. In the equation given below, x denotes voice signal, min denotes the minimum value in voice signal and max denotes the maximum value.

$$x' = \frac{x - \min}{\max - \min} \tag{11}$$

C# Programming Language was used for the speech recognition software developed in this study. The visual design of the speech recognition software developed is given in Figure 1. There are three menus in the software. There are two options within the voice menu. Calculated eigenvector and eigenfaces related to voice data recorded to database previously become ready to use with load voice data option. A new voice and data obtained with PCA algorithm from this related voice is added to database with add voice option.

The most appropriate word is determined by comparing results of the voice related to the word entered and the words stored in database obtained from PCA and also graphics and matching results of the word found out are displayed by using Analysis and Speech Recognition menus. Furthermore, time analysis values of the steps of the process in PCA algorithm are also displayed in the screen.

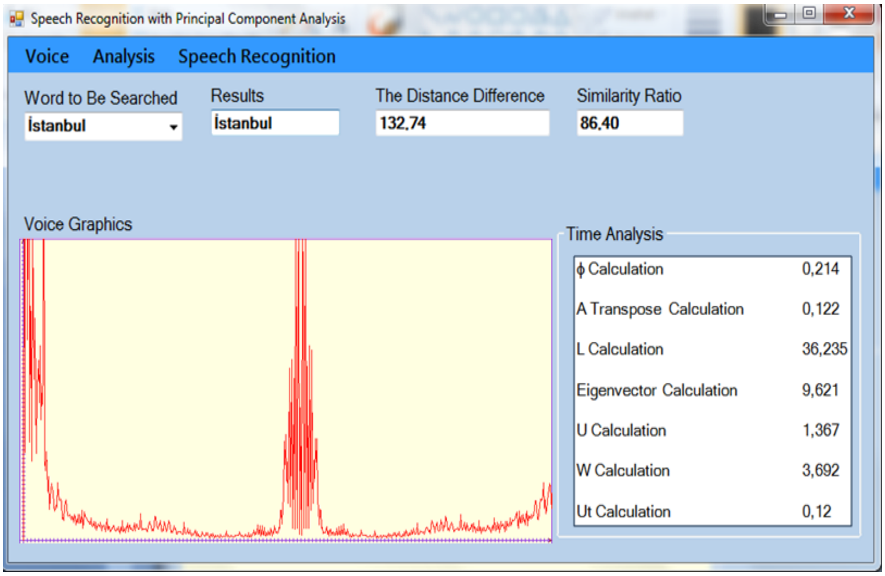


Fig. 1. Software Interface

PCA method is used in the speech recognition software developed. 100 different words and 3 different voice records related to these words were used in obtaining PCA. In other terms, PCA algorithm, eigenvalues and eigenfaces of totally 300 voice records were calculated. Before the application of PCA algorithm, determination of starting and end points and also normalisation pre-processes were applied to the voice records. The sizes of the voices pre-processed were minimised with PCA and were used in the word recognition process by calculating eigenvalues and eigenvectors.

The next step in speech recognition is the recognition of the new voice in which the voice is compared to the words in order to find out whether it is a known word or not. The weights are obtained with the application of PCA algorithm steps. Euclidean distance is calculated as given in equation 12 in order to find out the distances between the voices where Ω denotes $\Omega=[w_1,w_2,...w_M]$.

$$d_k = \|(\Omega - \Omega_k)\| \qquad (12)$$

Similarity ratio is calculated by dividing this value into the number of words [14].

4. Experimental Results

100 different words and a database consists of 300 different words which were vocalized with three different tones were used in this study. Samples of voice graphics related to the voice records obtained are given in Figure 2. Voice records were obtained with recording devices which have noise filtering features in special studio settings where the noises are minimised. Words forming the database are given in Table 1. When constituting word table, various grammatical features such as vowel harmonies, vowels and consonants, emphasis, stress, intonations, etc. were taken into consideration. When the similarity ratio was decided as 70% threshold value as a result of tests performed, it was found out that the word recognition ratio of the software was 86%. In other terms, the software recognised 86 words out of 100 words, the recognition process failed in the recognition of 14 words. Successful and unsuccessful sample results are given in Table 2. The durations of the operations seen in screenshot given in Figure 1 were obtained with laptop computer having Core i5 450M 2,4 GHz microprocessor, 4GB RAM and Windows 7 operating system. It was found out that the calculation of covariance matrix requires too much process time.

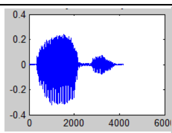
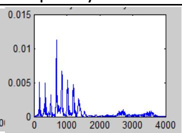
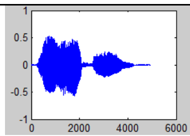
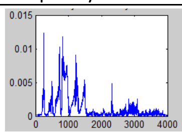
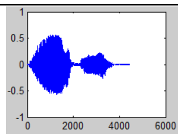
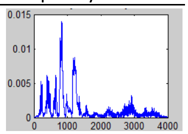
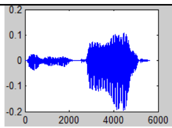
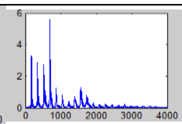
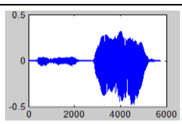
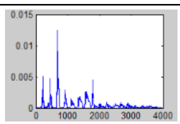
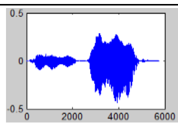
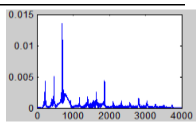
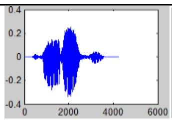
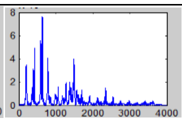
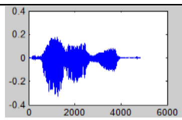
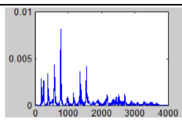
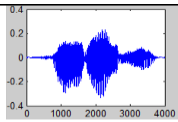
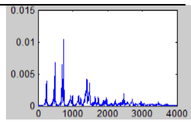
Words	Low Voice Level Record in Time Domain	Low Voice Level Record in Frequency Domain	Normal Voice Level Record in Time Domain	Normal Voice Level Record in Frequency Domain	High Voice Level Record in Time Domain	High Voice Level Record in Frequency Domain
İstanbul						
Ankara						
İzmir						

Fig. 2. Voice Graphics

Table1. Words

Abi	At	Can	El	Gitmek	İnsan	Kız	Olmak	Selam	Süleyman
Adam	Ay	Çocuk	Emine	Göz	İstanbul	Köpek	Onlar	Sen	Tamam
Akşam	Baba	Çok	Emrah	Gül	İzmir	Mavi	Otopark	Senin	Telefon
Amca	Barış	Dağ	Erik	Gün	Kalem	Nasıl	Özgürlük	Sepet	Teşekkür
Ana	Ben	Dayı	Erkek	Güneş	Kalp	Ne	Özlemek	Sevgi	Teyze
Ankara	Benim	Deniz	Esma	Hala	Kara	Neden	Para	Sevgili	Türk
Araba	Beyaz	Dere	Ev	Hayat	Karabük	Nokta	Renk	Sevmek	Yasemin
Arkadaş	Bilgisayar	Dost	Evet	Hayır	Kardeş	Odun	Sarı	Siyah	Yemek
Aşık	Bir	Dünya	Gazi	İki	Kırmızı	Öğretmen	Satmak	Şok	Yeşil
Aşiyan	Bu	Ekmek	Gece	İnek	Kitap	Okul	Saygı	Su	Yok

Table2. Sample Results

Words Searched	**Result**	**Calculated Distance**	**Recognition Rate (%)**
Renk	Word could not be found	15732,26	0,60
İzmir	İzmir	132,36	75,75
Olmak	Olmak	106,25	94,36
Özgürlük	Word could not be found	163,31	61,34
İstanbul	İstanbul	132,74	86,40

5. Conclusions

As a result speech recognition software based on PCA method was developed within this study. PCA method was preferred as a rapid solution approach since this method can easily be applicable. It was seen that fine results were obtained from clear records by utilizing speech recognition software proposed in this study. Furthermore, voice tone amplitude changes that do not affect the voice amplitude were also tolerated by the system. Time analysis values of the steps of process included in PCA algorithm were obtained, recognition success was displayed. The necessities of re-calculation of eigenvector and PCA for each word which will be added to the system generate a disadvantage.

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6. Biographies

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